

I'll break down the most important things I found, in plain language.

The Big Picture

You're building a tool to help emergency coordinators triage bushfire images quickly. The research shows four main problems these coordinators face right now — and your tool can directly address them.

Problem 1: Information Overload — They're Drowning in Data

What happens: When a major fire starts, coordinators get hundreds of images, alerts, and data feeds all at once. They're trying to look at maps, read reports, answer radio calls, and check multiple systems — all while making life-or-death decisions.

The real cost: A firefighting expert put it bluntly: "Every time we give a coordinator something else to monitor, we take their eyes off something else." Studies show this causes "tunnel vision" — they focus so hard on one thing they miss critical changes happening elsewhere.

How your tool helps: By automatically sorting images into severity levels and showing only the most urgent ones first, you're *reducing* what they need to hold in their head — not adding another feed to watch.

Problem 2: They Don't Trust Automated Rankings — Especially in Hard Cases

What happens: Researchers studied ambulance dispatchers who had an automatic assignment system. When situations were routine, dispatchers used the system. But as situations got *harder and more critical*, dispatchers *stopped trusting it* and went back to doing things manually instead.

Why it matters: This is exactly backwards from what we'd hope. The system should help most when it's hardest — but if it's not transparent, coordinators won't use it.

The real numbers: In complex situations (traffic accidents), dispatcher reliance fell by 41%. On the most critical cases, they abandoned the system altogether.

How your tool must work: Every ranking your AI makes needs to be explainable. A coordinator should be able to click on a severity level and see *why* the AI ranked it that way.

And they need to be able to override it easily in one action. If you hide the reasoning, they won't trust you when it matters most.

Problem 3: "Cry Wolf" Alert Fatigue — Too Many Alerts Make Them Ignore Everything

What's happening right now: Victoria's official bushfire app sends coordinators dozens of similar notifications on a bad fire day. Real users are complaining: "I get 13 almost identical notifications just to say conditions are dangerous. This teaches me to ignore everything from this app."

The psychological effect: When you get constant low-level alerts, your brain learns to dismiss them. Then when a *real* life-threatening alert comes, you ignore it because you've trained yourself not to pay attention.

An even bigger problem: The app doesn't distinguish between a single bush catching fire in the suburbs versus an actual community-level bushfire. Users report this is "dangerous" because it means a real emergency looks the same as a minor incident.

How your tool must work: Your four-level severity scale is directly solving this. You need to make sure that high-severity fires *look and feel* genuinely different from low-severity ones. Use colour, size, or visual weight to make that distinction obvious. If everything screams "urgent," nothing is actually urgent.

Problem 4: Map Clutter Hides What Matters

What happens: When coordinators look at a live map with dozens or hundreds of fire detections all overlapping, the symbols become a meaningless blur. They literally can't see what's important because there's too much visual noise.

Real-world example: Military and emergency map standards exist precisely because overlapping symbols on a shared map confuse people under pressure.

How your tool must work: When you're showing multiple fire detections on your live map, group them intelligently. Don't show every single pixel as a separate marker — cluster them, and only expand the detail when a coordinator zooms in. Make the visual symbols clear and standardised so everyone reads the map the same way.

Problem 5: AI Confidence Needs to Be Honest (Not Hidden or Overblown)

What's happening: When AI systems don't explain how confident they are, coordinators either over-trust them (and miss problems) or don't trust them at all (and ignore them).

A real example: California's ALERTCalifornia system shows a percentage confidence score alongside every fire detection — "47% confidence this is a fire." This transparency lets trained staff decide whether to investigate further. It works.

The trap to avoid: If your AI says "this is a high-severity fire" but doesn't show *how confident* it is, coordinators won't know if they should act immediately or check it themselves first.

How your tool must work: If the AI is uncertain about a classification (say, smoke makes it hard to see), show that explicitly. Route truly low-confidence cases to a "human review needed" queue instead of forcing the coordinator to trust a shaky call. This isn't a failure — it's the right thing to do.

Problem 6: Interoperability — Systems Don't Talk to Each Other

What's happening: Different agencies use different systems (VicEmergency, local dispatch software, weather apps, mapping tools). Coordinators have to manually copy information from one system to another, or worse, miss data because it's in a system they're not checking.

The cost: During the 2019–20 Black Summer fires, cross-border radio incompatibility was so bad a NSW helicopter had to land and hold an in-person meeting just to share situational awareness with Queensland ground crews. That cost critical time.

Why it matters for your tool: Your system should try to *ingest* data from multiple sources (different image formats, drone feeds, satellite data) without making the coordinator do manual work. And when you output results (the severity map, resource recommendations), they should be easy for other agencies to import into *their* systems.

The Core Design Principle

All of these problems trace back to one thing: **coordinators need to see what's most important, understand why the system thinks it's important, and be able to override it if they disagree** — all without drowning in noise or second-guessing themselves.

What This Means for Your Wireframes & Build

1. **Severity differentiation is not cosmetic** — use visual weight (colour, size) to make high-severity fires unmistakably different from low ones. Test with real people to make sure.
2. **Every AI decision needs a reason** — if you classify something as "high severity," show the coordinator what evidence led to that (e.g., "image clarity: high," "structure damage detected," "wind conditions severe").
3. **Low confidence = human review, not auto-classify** — if the AI isn't sure, send it to a queue for confirmation rather than forcing the coordinator to trust an uncertain call.
4. **Map clutter kills comprehension** — cluster detections by severity and location, use standardised symbols, and only show detail on demand.
5. **Make override easy** — if a coordinator wants to change the severity or priority, that should take one click, not five. If they can't easily override, they'll stop using the system.
6. **Show the reasoning behind prioritisation** — when locations are ranked by resource need, a coordinator should be able to click and see *why* location A is ranked before location B. If the ranking feels arbitrary, they won't trust it.

This is why you're building an "aid to decision-making," not an "automated decision." The coordinator stays in control; the AI just helps them see what matters fastest.

Sources:

Absolutely. Here are the direct sources for each major finding:

Problem 1: Information Overload

"Every time we give a coordinator something else to monitor..."

- Source: FireRescue1 article on firefighter technology adoption
- URL: <https://www.firerescue1.com/technology/from-skepticism-to-standard-how-firefighters-adopt-new-technology>

Tunnel vision and cognitive overload under pressure:

- Source: PNNL (Pacific Northwest National Laboratory) — "Attention and Situational Awareness in First Responder Operations"
 - Direct PDF: https://www.pnnl.gov/sites/default/files/media/file/RTA_Situational_Awareness.pdf
 - Also: Research showing "context switching" between systems reduces response speed by 15–25%
 - Source: Gartner 2024 IT Infrastructure & Operations report (cited in VUWall blog)
 - URL: <https://vuwall.com/blog/2025/control-what-you-see-smarter-simpler-control-rooms/>
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Problem 2: Ambulance Dispatcher Study — Trust Collapses on Hard Cases

The 540,033-event study showing reliance falls on complex/critical cases:

- Source: Moshe, Meyer, Jaffe & Bitan (2026), *International Journal of Human-Computer Interaction*
- Title: "Between Compliance and Complexity: Dispatchers' Use of an Automatic Emergency Vehicle Assignment System"
- DOI: <https://doi.org/10.1080/10447318.2025.2607554>
- Full text (Taylor & Francis): <https://www.tandfonline.com/doi/full/10.1080/10447318.2025.2607554>
- PDF: <https://www.tandfonline.com/doi/pdf/10.1080/10447318.2025.2607554>

Key finding: 41% drop in automation reliance for traffic accidents (complex cases); reliance declined from 31.9% (2018) to lower in 2019.

Problem 3: "Cry Wolf" Alert Fatigue — VicEmergency App

Real user complaints about too many notifications:

- Source: VicEmergency App Store reviews (Apple App Store)
- URL: <https://apps.apple.com/au/app/vicemergency/id356559665?see-all=reviews&platform=iphone>
- Direct quote: "I get no less than 13 almost identical notifications just to inform me that conditions are dangerous. This is a ridiculous level of notifications."

Users reporting lack of severity differentiation:

- Same source (App Store reviews)
- Quote: "There is no distinction in the app between a life-threatening bushfire and a single bush that's caught fire in the suburbs somewhere."

Academic literature on warning fatigue:

- Sutton & Wood (2025), *Journal of Contingencies and Crisis Management*
- Title: "Opting Out: Over-Alerting and Warning Fatigue in the Era of Wireless Emergency Alerts"
- Source: University at Albany Scholars Archive
- URL: https://scholarsarchive.library.albany.edu/ehc_fac_scholar/20/

Australian bushfire context specifically:

- Mackie (2013): "Warning Fatigue: Insights From the Australian Bushfire Context"
 - (Reference in research but direct URL not readily available; cite the Sutton & Wood paper as primary source)
-

Problem 4: Map Clutter and Symbol Standardisation

HCI research on emergency map symbols and clutter reduction:

- Source: Opach & Rød (2022), *Cartography and Geographic Information Science*
- Title: "A user-centric optimization of emergency map symbols to facilitate common operational picture"
- URL: <https://www.tandfonline.com/doi/full/10.1080/15230406.2021.1994469>

NATO APP-6 symbology and clutter:

- Mentioned in the above paper
- Also referenced in: NAPSG Foundation (DHS S&T/FEMA-backed) emergency map symbology standards

Common Operating Picture (COP) design principles:

- Source: Resgrid Blog (2026)
- Title: "Common Operational Picture: Guide for First Responders 2026"
- URL: <https://blog.resgrid.com/common-operational-picture/>
- Practical guidance: "Build the display around decisions, not around available widgets."

Cognitive overload in control rooms:

- Source: Activu Corporation (control-room design specialists)
 - URL: <https://www.activu.com/eoc-the-definitive-guide-to-emergency-operations-center-management/>
 - Quote: "Modern emergency management often suffers from a paradox of information. While agencies have access to more data than ever before, this volume frequently creates 'screen fatigue' rather than clarity."
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Problem 5: AI Confidence and Uncertainty Handling

Mis-calibration of AI confidence — experimental study:

- Source: arXiv paper (2024)
- Title: "Understanding the Effects of Miscalibrated AI Confidence on User Trust"
- (arXiv ID available in research; direct URL: [search arxiv.org](https://arxiv.org) for the title)

Optimal confidence threshold for AI/human delegation:

- Source: IJACSA (International Journal of Advanced Computer Science and Applications)
- Title: "Confidence-Based Trust Calibration in Human-AI Teams"

Uncertainty visualization on maps — time-pressure effects:

- Source: Korporaal, Ruginski & Fabrikant (2020), *Frontiers in Computer Science*
- Title: "Effects of Uncertainty Visualization on Map-Based Decision Making Under Time Pressure"
- URL: <https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2020.00032/full>

MacEachren et al. on uncertainty visualization in geospatial data:

- Source: "Visualizing Geospatial Information Uncertainty: What We Know and What We Need to Know"
- URL: https://www.academia.edu/320972/Visualizing_Geospatial_Information_Uncertainty_What_We_Know_and_What_We_Need_to_Know

ALERTCalifornia — real-world confidence display example:

- Source: UC San Diego Supercomputer Center / CAL FIRE partnership
 - Reference: Fire Safe Marin (agency reporting)
 - Quote about confidence percentage display: "That dot .47 is telling me that (it has) a less than 50% chance of thinking it's a fire."
 - Note: adoption figures ("over 1,200 fires detected," "beating 911 over 30%") are from agency/press reports, not independent peer review
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Problem 6: Interoperability & Data Fragmentation

FIRIS (California Fire Response Information System) documented failures:

- Source: California Fire Service Leadership Alliance
- Title: "FIRIS White Paper" (2024)
- URL: https://cdn.ymaws.com/www.calchiefs.org/resource/resmgr/documents/whitepapers/FIRIS_White_Paper.pdf
- Key issue: "Single-channel access for most Slack chat users limits the information distributed ... having limited access to channels has unforeseen consequences that hamper situational awareness and firefighter safety."
- Also distinguishes "information" (raw data) vs "intelligence" (analysed data)

Black Summer bushfires — no national data platform:

- Source: Royal Commission into National Natural Disaster Arrangements (Australian Government, October 2020)
- URL: <https://www.royalcommission.gov.au/natural-disasters/report>
- Finding: "Australia does not have a nationally consistent bushfire information system"

Cross-border radio incompatibility during Black Summer:

- Source: ArcGIS StoryMap on Australia's Black Summer
- URL: <https://storymaps.arcgis.com/stories/bd6c8f95d39d424187fc45258006ebbd>
- Reference also in: AirMed&Rescue article
- URL: <https://www.airmedandrescue.com/latest/long-read/royal-commission-review-australias-black-summer>

Royal Commission recommendation for common platforms:

- Source: Royal Commission into National Natural Disaster Arrangements — List of Recommendations
- URL: <https://naturaldisaster.royalcommission.gov.au/publications/html-report/list-of-recommendations>
- Recommendation 4.1: "Australian, state and territory governments should create common information platforms and share technologies to enable collaboration in the production, analysis, access, and exchange of information, data and knowledge."

General emergency-services interoperability problems:

- Source: UK multi-agency inquiry findings (Manchester Arena, Grenfell Tower)
 - Reference: 2025 Safety Science study on UK emergency-service interoperability
 - Also: Joint Emergency Services Interoperability Principles (JESIP) failures documented in Manchester Arena Inquiry Vol. 2 (November 2022)
-

AI-Specific Findings: Four-Level Severity Scale & Human-in-the-Loop

xView2 dataset — four-level Joint Damage Scale precedent:

- Source: Defense Innovation Unit / xBD dataset
- Paper: Gupta et al., arXiv:1911.09296
- Title: "xView2: Multimodal Multi-Task Convolutional Neural Networks for Rapid Building Damage Assessment after Disasters"
- Dataset covers 850,736 annotated buildings; deployed on 2020 California wildfires and 2019–20 Australian bushfires
- Levels: No Damage (0), Minor Damage (1), Major Damage (2), Destroyed (3)
- Grounded in: HAZUS, FEMA Damage Assessment Manual, Kelman scale, EMS-98 scale
- CMU Software Engineering Institute reference: https://www.sei.cmu.edu/our-work/projects/display.cfm?customel_datapageid_4050=295280

Black-box opacity as adoption barrier:

- Source: MDPI *AI* (2025, Vol. 6, Issue 10, p. 253)
- Title: "AI for Wildfire Management"
- Quote: "Despite increasingly sophisticated research, the operational use of AI in wildfire contexts remains limited ... AI models offer higher accuracy but are limited by interpretability issues."
- Published: October 1, 2025
- URL: <https://www.mdpi.com/2413-8460/6/10/253>

Human-in-the-loop case study — Rolling Fork tornado (2023):

- Source: arXiv (2023)
 - Title: "Rapid building damage assessment workflow: An implementation for the 2023 Rolling Fork, Mississippi tornado event"
 - American Red Cross partnership
 - Achieved: precision 0.86 / recall 0.80, executed end-to-end in under 2 hours per satellite scene
 - URL: <https://arxiv.org/pdf/2306.12589>
-

Automation Bias & Reliance in High-Stakes Situations

Automation bias in time-critical systems:

- Source: Maritime Safety Innovation Lab
- Title: "Automation Bias in Intelligent Time-Critical Decision Support Systems"
- PDF: <https://maritimesafetyinnovationlab.org/wp-content/uploads/2023/02/Automation-Bias-in-Intelligent-Time-Critical-Decision-Support-Systems.pdf>
- Key finding: 40% of pilots "reasoned less or none at all when confronted with uncertainty"

Triage tool design failure — Lille pediatric ED:

- Source: JMIR Human Factors (2020)
- Title: "Applying a Human-Centered Design to Develop a Patient Prioritization Tool for a Pediatric Emergency Department"
- URL: <https://humanfactors.jmir.org/2020/3/e18427/>
- Finding: Electronic triage systems fail when algorithms don't match clinicians' actual decision-making strategies

Summary Table of Sources

Finding	Source Type	Direct Link
Information overload	Trade article + Government lab report	https://www.firerescue1.com/technology/... ; https://www.pnnl.gov/sites/default/files/media/file/RTA_Situational_Awareness.pdf
Dispatcher trust collapses on hard cases	Peer-reviewed journal (2026)	https://doi.org/10.1080/10447318.2025.2607554
VicEmergency alert fatigue	App Store user reviews	https://apps.apple.com/au/app/vicemergency/id356559665?see-all=reviews
Map clutter/symbols	Peer-reviewed journal (2022)	https://www.tandfonline.com/doi/full/10.1080/15230406.2021.1994469

Finding	Source Type	Direct Link
AI confidence calibration	Academic journals + arXiv	arxiv.org (search title); IJACSA journal
Uncertainty visualization	Peer-reviewed journal (2020)	https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2020.00032/full
FIRIS interoperability	White paper (2024)	https://cdn.ymaws.com/www.calchiefs.org/resource/resmgr/documents/whitepapers/FIRIS_White_Paper.pdf
Royal Commission findings	Government inquiry (2020)	https://www.royalcommission.gov.au/natural-disasters/report
xView2 four-level scale	arXiv + Defense Innovation Unit	arXiv:1911.09296; https://www.sei.cmu.edu/our-work/projects/...
AI interpretability barrier	Peer-reviewed journal (2025)	https://www.mdpi.com/2413-8460/6/10/253
Human-in-the-loop precedent	arXiv case study (2023)	https://arxiv.org/pdf/2306.12589
Automation bias	Maritime research paper	https://maritimesafetyinnovationlab.org/wp-content/uploads/2023/02/Automation-Bias-in-Intelligent-Time-Critical-Decision-Support-Systems.pdf

Caveats:

- Some arXiv papers don't have stable DOI URLs; search the title directly on arxiv.org
- ALERTCalifornia adoption figures come from agency/press reports, not peer-reviewed studies — treat as case evidence, not definitive proof
- Some sources are very recent (2026) — they are news/agency reports rather than peer review
- App Store reviews are user-generated; they're real feedback but not academic data

Does this help? Let me know if you need the full text of any of these, or if you want me to explain how to access paywalled papers.